

E-Commerce Sales Analytics Project

Overview

This project analyzes 50,000 e-commerce transactions using MySQL and Power BI. The objective is to transform raw transactional data into meaningful business insights through database design, SQL analytics, and interactive dashboards.

Tools Used

- MySQL
 - MySQL Workbench
 - SQL
 - Power BI
 - GitHub
-

Dataset Description

The dataset contains:

- Transaction ID
- Customer Name
- Age
- Country
- Product Category
- Purchase Amount
- Payment Method
- Transaction Date

Total Records: 50,000

Database Design

The raw dataset was normalized into three primary tables:

Customers

Stores customer information.

Categories

Stores unique product categories.

Transactions

Stores transaction-level records linked to customers and categories using foreign keys.

Database Relationships

Customers (1) → Transactions (*)

Categories (1) → Transactions (*)

This structure follows a star schema model suitable for business intelligence reporting.

SQL Concepts Implemented

Basic SQL

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING

Intermediate SQL

- JOINS
- Aggregate Functions
- Subqueries

Advanced SQL

- Window Functions
- RANK()

- DENSE_RANK()
 - ROW_NUMBER()
 - Common Table Expressions (CTE)
 - Views
 - Stored Procedures
-

Business Analysis Performed

Revenue Analysis

Calculated total revenue generated by all transactions.

Customer Analysis

Identified top spending customers and customer purchasing behavior.

Category Analysis

Determined highest-performing product categories.

Country Analysis

Compared revenue generation across countries.

Payment Analysis

Analyzed customer payment preferences.

Monthly Trend Analysis

Tracked revenue performance over time.

Power BI Dashboard

The dashboard includes:

- Total Revenue KPI
- Total Transactions KPI
- Average Order Value KPI
- Total Customers KPI

- Revenue by Category
 - Monthly Revenue Trend
 - Revenue by Country
 - Payment Method Analysis
 - Top Customers Table
 - Interactive Filters
-

Key Learnings

- Relational Database Design
 - Data Normalization
 - SQL Analytics
 - Business Intelligence
 - Dashboard Development
 - Data Visualization
 - KPI Reporting
-

Conclusion

This project demonstrates an end-to-end analytics workflow starting from raw data ingestion and database design to advanced SQL analysis and Power BI dashboard development. It showcases practical skills in data analytics, business intelligence, and reporting.